SEQUENCE 10

# Variations and extrema of a function

1–12 February 2027

## 10.1 Increasing, decreasing and monotonicity

**Key point · Definitions.** On an interval $I$, $f$ is increasing if for all $a, b \in I$, $a < b \Rightarrow f(a) \leq f(b)$; decreasing if $a < b \Rightarrow f(a) \geq f(b)$. The variations are strict when the inequalities between images are strict. A constant function has the same value everywhere; it is both increasing and decreasing in the broad sense. A function that is monotonic on $I$ is increasing or decreasing there.

**Method · Table of variations.** Indicate the domain, the intervals of monotonicity, the values at the endpoints when defined, then the arrows. A sign table shows whether $f(x)$ is positive or negative; a table of variations shows how its values change.

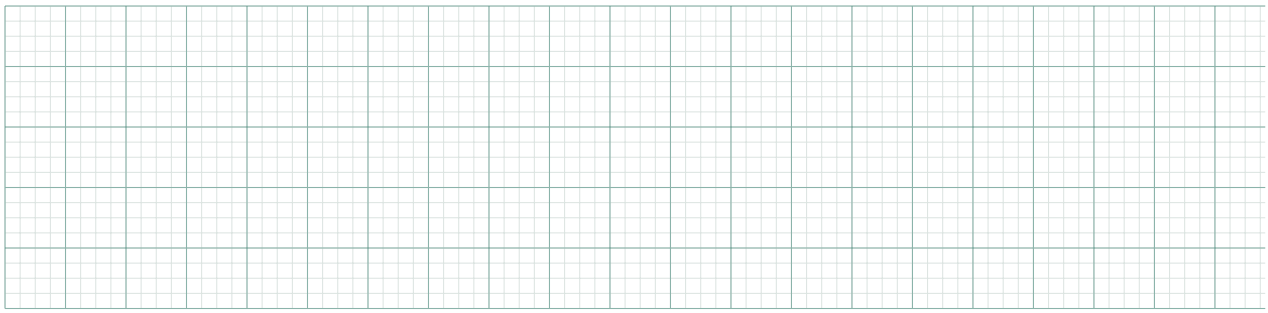
**Example 1.** On $[-3, 4]$, $f(x) = (x-1)^2 - 2$. Using the square, extending the curve studied in sequence 6, draw its table of variations. Give two numbers $a < b$ such that $f(a) > f(b)$.

**Example 2.** Can a function be negative and increasing? Give an example on an interval.

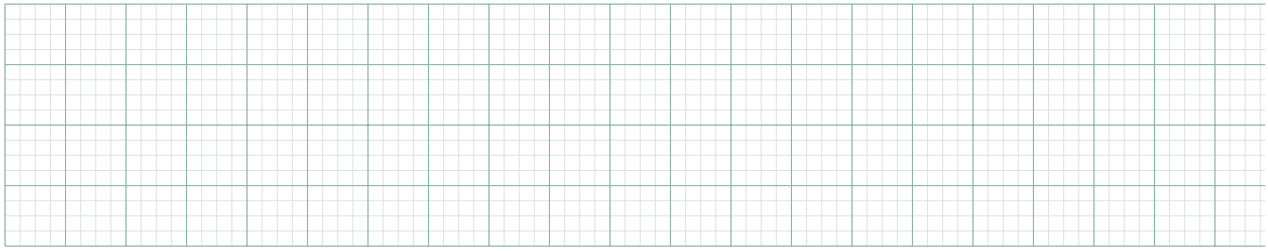
## 10.2 Maximum and minimum on an interval

**Key point · Definitions.** $f$ has a maximum at $a \in I$ if, for every $x \in I$, $f(x) \leq f(a)$. The value of the maximum is $f(a)$. $f$ has a minimum at $b \in I$ if, for every $x \in I$, $f(x) \geq f(b)$. An extremum must be attained. Distinguish its value from the abscissa where it is attained.

**Example 3.** Return to $f(x) = (x-1)^2 - 2$ on $[-3, 4]$. Give the minimum and maximum, then where they are attained. Repeat on $[0, 4]$.

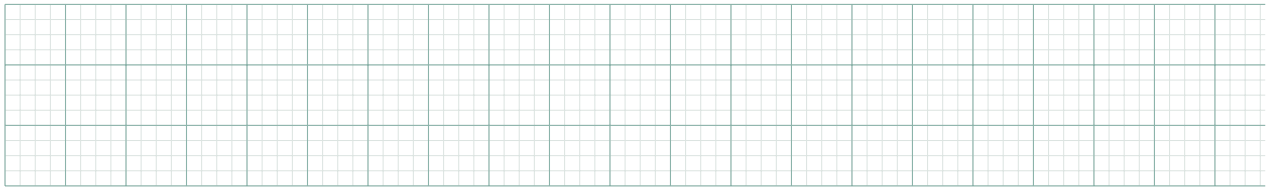


**Example 4.** Does the function $f(x) = x$ have a maximum on $[0, 1)$? A minimum?



**Key point · Reading without inventing.** A table of variations does not provide all the intermediate images. It allows comparing certain images; if the abscissae lie on different branches, the comparison may be indeterminate.

**Example 5.** Does the mere fact that a function decreases then increases allow comparing two images on either side of the minimum?



## 10.3 Variations of affine functions

**Key point · Property and rate of change.** For $f(x) = mx + p$ and $a \neq b$,

$$\frac{f(b) - f(a)}{b - a} = m.$$

If $m > 0$, $f$ is strictly increasing; if $m < 0$, strictly decreasing; if $m = 0$, constant.

**Proof to write.**

**Example 6.** Determine the variations of $f(x) = 3x + 2$, $g(x) = 7 - 6x$, $h(x) = -x$. Give the sign of $2x - 6$ and explain why the sign of this function does not stay constant although it is increasing.

**Example 7.** An affine function $u$ satisfies $u(2) = 5$, $u(6) = 13$. Determine $u$, then its minimum and maximum on $[-1, 4]$.

## 10.4 Variations of the square and the absolute value

**Key point · Properties.** $x \mapsto x^2$ and $x \mapsto |x|$ are strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, +\infty)$. Their minimum on $\mathbb{R}$ is $0$, attained only at $0$.

**Proof to write.**

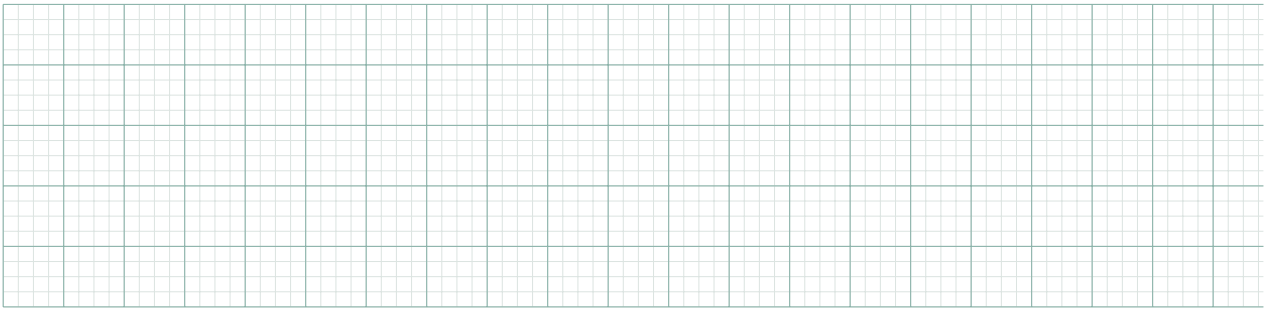
**Example 8.** Compare without a calculator: $(-3.2)^2$ and $(-3.1)^2$; $1.2^2$ and $1.3^2$; $|-5|$ and $|-2|$.

**Example 9.** Bound $x^2$ if $-3 \leq x \leq 2$. Why can we not simply square the two bounds?

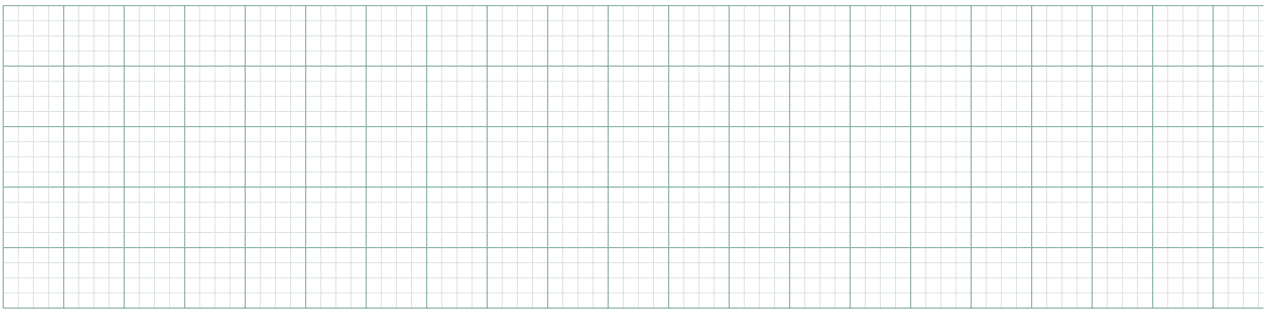
## 10.5 Variations of the inverse function

**Key point · Property.** The inverse function is strictly decreasing on $(-\infty, 0)$ and on $(0, +\infty)$, separately. It is not decreasing on the whole of $\mathbb{R}^*$.

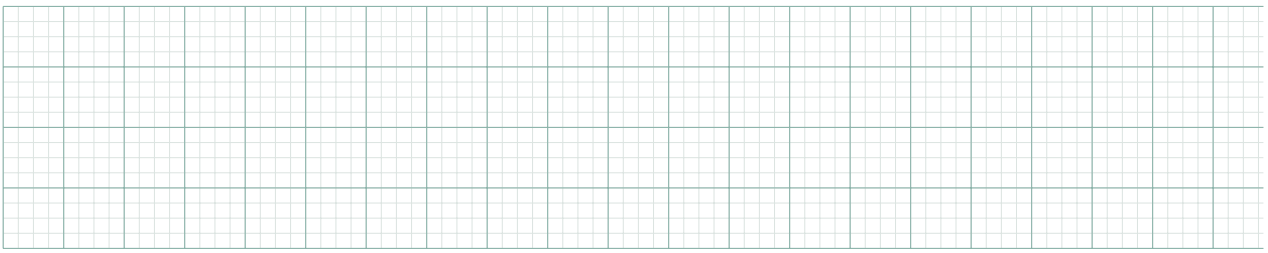
**Proof to write.**



**Example 10.** Compare $1/(-4)$ and $1/(-2)$, then $1/3$ and $1/5$. Bound $1/x$ if $2 \leq x \leq 5$, then if $-5 \leq x \leq -2$.



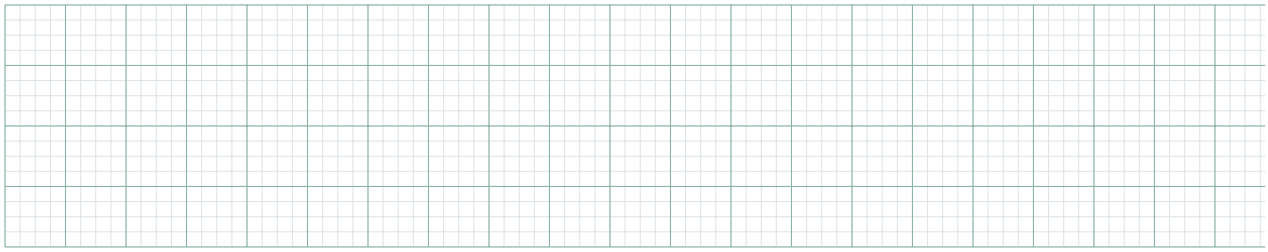
**Example 11.** Why is the interval $[-1, 1]$ not suitable for directly applying a variation rule of the inverse function?



## 10.6 Relative position of a number and its square

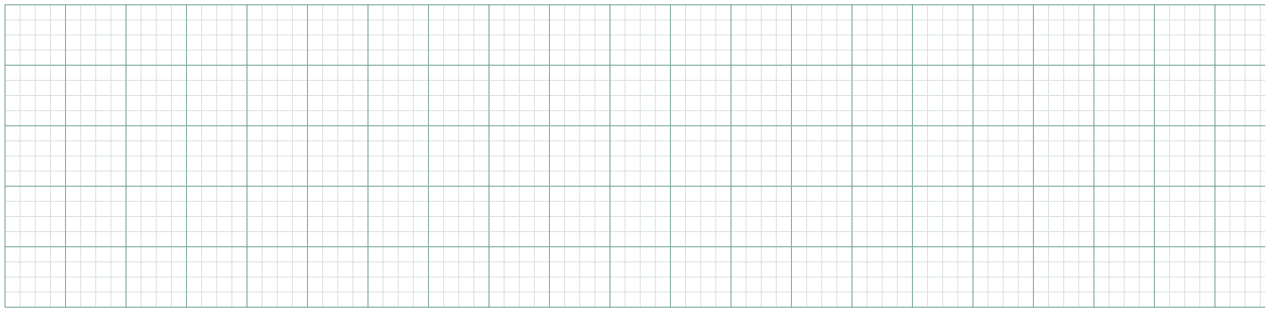
**Key point · Property on the positives.** For $0 \leq x \leq 1$, $x^2 \leq x$; for $x \geq 1$, $x^2 \geq x$. Equality holds at 0 and 1.

**Proof to write.**

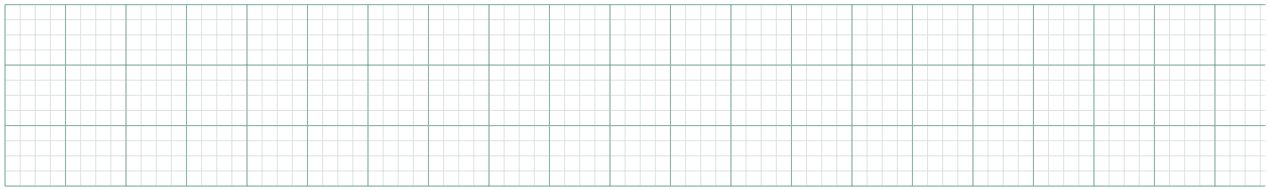


**Key point · Square root and cube.** The square root function is strictly increasing on $[0, +\infty)$. The cube function is strictly increasing on $\mathbb{R}$.

**Example 12.** Explain why the square root is increasing from the increase of the square on the positives. For the cube, use $b^3 - a^3 = (b-a)(a^2+ab+b^2)$.



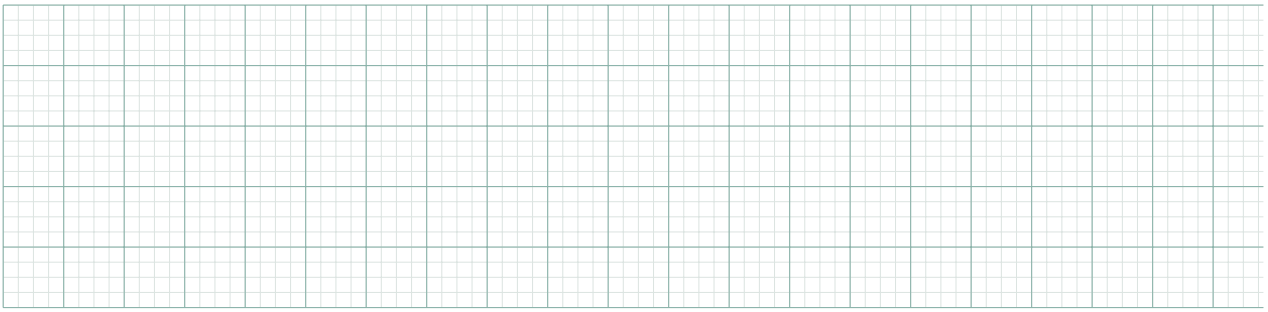
**Example 13.** Compare $\sqrt{5}$ and $\sqrt{6}$, then $(-2)^3$ and $(-1)^3$.



## 10.7 Optimising: conjecture and prove

**Example 14.** A rectangle of perimeter 20 m has area $A(x) = x(10 - x)$ with $x \in (0, 10)$.

1. Conjecture the maximum area using a table of values.
2. Show that $A(x) = 25 - (x - 5)^2$. Deduce the exact maximum and the corresponding dimensions.



```
def aire(x):
    return x*(10-x)

def maximum_grille(N):
    # N entier >= 2 ; points strictement entre 0 et 10
    meilleur_x = 10/N
    meilleur_y = aire(meilleur_x)
    for k in range(2, N):
        x = 10*k/N
        if aire(x) > meilleur_y:
            meilleur_x, meilleur_y = x, aire(x)
    return meilleur_x, meilleur_y
```

**Example 15.** What does the program return for $N = 10$? Does a numerical sweep always prove the maximum of a function on an interval?

